Thermal Expansion

Name _____

Station 1
Place your hands on the flask and record what you observe.

Explain your observation.

_____

Station 2
Place your hands on the flask and record what you observe.

Explain your observation.

_____

Station 3
Attempt to pass the ball through the ring. Does it fit through the ring?
Answer _____

Now heat the ball for 1 minute in a bunsen burner flame.
Attempt to pass the ball through the ring. Does it fit through the ring?
Answer _____

Explain the difference between your two answers.

_____

Station 4
Attempt to fit the bar in the gauge. Does it fit?
Answer _____

Now heat the bar in a Bunsen burner flame for 2 minutes.
Attempt to fit the bar in the gauge. Does it fit?
Answer _____

Explain the difference between your two answers.

You can now answer some general questions about the expansion and contraction of solids liquids and gases when they are heated and cooled.

Q1 Do all materials expand and contract at the same rate when heated and cooled?

Q2 Rank solids, liquids and gases in order from the one which expands least when heated to the one which expands most when heated.

Q3 A cheap pendulum clock is seen to run slow on a very hot day. Explain why the heat may cause this .

Q4 Steel frames used in the construction of a factory unit buckle when the factory is accidentally involved in a damaging fire. Why may this happen?

Q5 One of the exceptions to the general rules about expansion and contraction is in the freezing of water. Find out about the *anomalous expansion of water* and explain why this occurs.

Extension

Q6 What is a thermostat?

Q7 Find out about a thermostat which makes use of a bimetallic strip to operate. Explain the big idea which makes the strip useful in the function of a thermostat.

Q8 Design a device which will turn on a lamp when a bimetallic strip is heated and will turn off the lamp when the bimetallic strip is cooled. Draw a diagram of the device below.